Practice Set 4 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Discuss in brief, present concepts in various GU cancers

Type: LAQ (25 marks)

Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.

Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).
- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive**

disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.

- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **^{177}Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.
- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

Diagram to draw: six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

Key points to remember:

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Q2. Robots in urology

Type: LAQ (25 marks) (also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP in **2000**; first paediatric robotic procedure **2002**.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) magnified view, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
S (2006) / Si	High definition, fourth arm, dual console for training
Xi (2014)	8-mm camera in any port, finer arms, docks from either side, no stirrups
SP (2018)	All instruments through one 2.5-cm cannula

- After patent expiry, other platforms appeared (e.g. **Hugo, Versius**, the Indian **SSI Mantra**).

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, less blood loss, shorter stay; equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable.
- **Retzius-sparing** and **hood** techniques preserve continence structures and give earlier continence.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy: shorter warm ischaemia and fewer conversions, so more complex and hilar tumours are done minimally invasively.

- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), less blood loss and transfusion.
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications.

4. Other procedures

Procedure	Why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children (small working space)
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery raises intracranial and intraocular pressure.

Complications specific to robotics

Problem	Details
Steep Trendelenburg	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position
Thermal bowel injury	Insulation failure, capacitive coupling; presents late
Robot malfunction	<1%; half found before the patient enters
Port-site hernia	Close larger port sites

- **Prevention**: pad pressure points, secure arms before tilting, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulators, animal and cadaver labs, then proctored cases. **FRS (Fundamentals of Robotic Surgery)** curriculum. Outcomes improve with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency).
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement.

> **Key points to remember:**
>
> - da Vinci 2000; Xi 2014; SP 2018; 7 degrees of freedom.
> - Yaxley: RARP equal to open at 12 weeks; RAZOR non-inferior; iROC more days out of hospital.
> - RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
> - Trendelenburg causes oedema and raised IOP; avoid shoulder braces.

Q3a. Female kidney donors: enumerate special precautions

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 4, Q3d)

Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.
3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

After donation

- **Defer pregnancy** until recovery and stable function (often 6–12 months).
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

Diagram to draw: flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

> **Key points to remember:**
> - Ensure free consent (Indian gender gap).
> - Pregnancy test; contraception; stop oestrogens.
> - Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
> - Defer pregnancy; high-risk antenatal follow-up.

Q3b. Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys.

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3e; 2016-Summer Paper 3, Q3f)

"Preemptive" is read as pre-implantation (procurement) biopsy.

Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- Still better survival than dialysis.

Categories

- **ECD (expanded criteria donor), UNOS (United Network for Organ Sharing):**
  - Age **60 or over**, or
  - Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.
  - Graft-loss risk about 1.7 times that of a standard donor.
- **KDPI (Kidney Donor Profile Index) >85%** replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Others:** DCD (donation after circulatory death); donor AKI (acute kidney injury); HCV-positive donors; older or hypertensive living donors.

Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components, each 0–3: total 0–12.**

Component	0	1	2	3
Glomerular global sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial and arteriolar narrowing	None	Wall thickness < lumen diameter	Wall = or slightly > lumen	Wall far > lumen, severe narrowing or occlusion

Allocation by score

Total	Grade	Use
0–3	Mild	**Single** kidney transplant
4–6	Moderate	**Dual** kidney transplant
7–12	Severe	**Discard**

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

Optimising marginal kidneys

- Short cold ischaemia; **HMP (hypothermic machine perfusion)** reduces DGF; old-for-old matching; informed consent.

Diagram to draw: Remuzzi grid with four compartments scored 0–3.

> **Key points to remember:**
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
> - Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
> - 0–3 single, 4–6 dual, ≥7 discard.

Q3c. Studer Neobladder.

Type: SAQ (10 marks) (also answered in 2015-Summer Paper 4, Q3e; selection and general points: see Practice Set 4 Paper 2, Q3e)

Principles

- **Studer (1989)**: detubularised ileal **orthotopic** reservoir with a long **isoperistaltic afferent limb**, joined to the urethra.
- **Detubularisation** stops coordinated contractions, so pressure stays low; a **sphere** (Laplace) gives most volume.
- The afferent limb damps reflux during Valsalva voiding and accepts **short ureters**. No staples in the reservoir. Continence depends on the **rhabdosphincter**.

Operative technique

1. **Isolate 54–56 cm of terminal ileum**, ending **15–20 cm proximal to the ICV (ileocaecal valve)**; restore bowel continuity.
2. **Reservoir: distal 40–44 cm** (two 20–22-cm limbs) opened on the antimesenteric border; sew the medial edges into a **U** (back wall).
3. **Afferent limb:** proximal tubular segment, **about 15 cm** (original 20 cm; 10–12 cm in later figures); close its end.
4. **Fold the U in the opposite direction** and close the anterior wall into a sphere.
5. **Ureters:** spatulated **end-to-side, refluxing** anastomoses to the afferent limb, interrupted 4-0, **stented**.
6. **Urethral anastomosis** at the most dependent point with pre-placed sutures; catheter and suprapubic tube.

- **Intracorporeal robotic Studer** follows the same steps.

Postoperative care and voiding training

- Mucus irrigation; oral bicarbonate; cystogram before catheter removal.
- Void **sitting**, relax the pelvic floor, strain gently; **timed voiding** every 2 hours, extended to 3–4 hours as capacity grows (under 200 mL to 500–600 mL in a year). **Night alarm.**

Outcomes

- Day continence 90–98%; night continence lower (up to half leak).
- Reflux on cystogram, yet kidneys preserved in about 95%; ureteroileal stricture **<3%**. **USC (University of Southern California) randomised trial:** antireflux T pouch gave no renal benefit and more reoperations.
- Late: pyelonephritis (commonest), acidosis, B12 deficiency, stones, retention.
- **Versus Hautmann W** (about 70 cm): smaller early capacity, less late retention.

Diagram to draw: marked ileal segment (afferent limb plus 40–44-cm U), then the folded sphere.

> **Key points to remember:**
> - 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U plus afferent limb.
> - Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%.
> - Sitting Valsalva voiding, timed voiding, night alarm.

Q3d. Neobladder.

Type: SAQ (10 marks)

See Practice Set 4 Paper 2, Q3e (full answer). Studer technique: see Q3c above.

Key points to remember (SAQ angle):

- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Needs a tumour-free urethral margin, eGFR (estimated glomerular filtration rate) >35–40 mL/min and ability to self-catheterise.
- Day continence about 90%; night incontinence up to half; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).

Q3e. Secondary tumors after urinary diversion.

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 4, Q3e)

Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more). Distinct from recurrent urothelial cancer.

Risk by diversion (German registry, Kälble 2011)

Diversion	Risk	Surveillance
Ureterosigmoidostomy	2.6% (6–29% in long-term series)	Regular colonoscopy **from 5 years**
Augmentation cystoplasty	1.6%	Endoscopy, often from 10 years
Colonic neobladder	1.3%	—
Ileocaecal pouch	0.14%	If symptomatic
Ileal neobladder	0.05%	No routine
Ileal conduit	0.02%	No routine

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis.
- **N-nitrosamines** from bacterial action on urinary nitrates (probably enhancers, not sole cause); chronic inflammation; growth factors (EGF, epidermal growth factor); ornithine decarboxylase; abnormal mucin.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

Pathology

- Mostly **adenocarcinoma**; also adenomatous polyps, urothelial carcinoma, sarcoma.
- **Augmentation:** adenocarcinoma and aggressive urothelial carcinoma, often advanced at diagnosis; earliest at 4 years. **Gastrocystoplasty** carries relatively high risk. Exstrophy and immunosuppression add risk.
- **Nephrogenic adenoma:** benign; commonest tumour after cystoplasty.

Presentation

- Haematuria, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent infection, mass.

Management

- Biopsy and staging.
- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total cystectomy with new diversion.
- **Prevention:** prefer ileal segments; counsel patients; lifelong follow-up.

Also know: after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance.

Diagram to draw: ureterosigmoid anastomosis with tumour at the urothelial–colonic junction.

Key points to remember:

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis.
- Excise defunctioned anastomoses.

Q3f. Urethral stents.

Type: SAQ (10 marks) (also answered in 2015-Summer Paper 4, Q3d)

Definition

- An endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.

Classification

Type	Examples	Features
Temporary	**Memokath** (nitinol), **Allium**	Designed to avoid epithelial incorporation; removable; left 6–12 months
Permanent, epithelialising	**UroLume**	Urothelium grows over it; **withdrawn**
Biodegradable	Experimental	Dissolve

Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate).
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: once an alternative to sphincterotomy; now **abandoned** for complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed and softens when cooled for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

Results

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Removable bulbar stents succeed only in small series. UroLume worked early but failed long term from ingrowth, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

Guidelines

- **AUA (American Urological Association)**: **no stent if urethroplasty is considered**; stents make reconstruction harder.

Recent advances

- **Optilume** paclitaxel balloon (FDA (Food and Drug Administration) approval 2023 for anterior stricture): **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind**: nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.

Also know: a soft silicone "urethral stent" (splint) after hypospadias repair or urethroplasty is a different device; in complicated hypospadias repairs it reduces meatal stenosis and fistula.

Diagram to draw: sagittal urethra with prostatic stent and bulbar stent distal to sphincter.

Key points to remember:

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Rapid Revision Sheet – Practice Set 4 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
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- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

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